

QUE : Explain the translation attribute of fields and its impact at the SQL level.

When you define a field with **translation=True**, Odoo stores translations in a separate table (**ir_translation**). Instead of saving multiple translations in the original table, Odoo retrieves the appropriate translation dynamically when a user views the record in their selected language.

Example: Defining a Translatable Field

```
from odoo import models, fields

class ProductTemplate(models.Model):
    _inherit = "product.template"

    description = fields.Text(string="Description",
translation=True)
```

- Here, the **description** field is **translatable**.
 - When a user switches the language, Odoo retrieves the corresponding translation from the **ir_translation** table.
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2. SQL-Level Impact of Translatable Fields

At the **SQL level**, Odoo does **not** store translations in the main table (**product_template** in this case). Instead, it saves them in the **ir_translation** table.

Structure of the **ir_translation** Table

When **translation=True** is enabled, Odoo creates entries in the **ir_translation** table with the following structure:

i d	name	lang	res_i d	src	value	typ e
1	product.template,field:de scription	fr_F R	10	"Red Shirt"	"Chemise Rouge"	fiel d
2	product.template,field:de scription	es_E S	10	"Red Shirt"	"Camisa Roja"	fiel d

Field Explanation:

- **name** : Specifies the model and field (product.template, field:description).
- **lang** : Stores the language (e.g., fr_FR, es_ES).
- **res_id** : Refers to the **ID of the record** (e.g., 10 for a product).
- **src** : The original text (source).
- **value** : The translated text.
- **type** : Specifies that this is a field translation.

3. How Odoo Uses Translations in SQL Queries

When fetching records, Odoo **joins** the **ir_translation** table dynamically based on the user's language.

Example Query in SQL

If a user views a product description in French (fr_FR), Odoo executes a query like this:

```
SELECT pt.id,  
       COALESCE(it.value, pt.description) AS  
description  
FROM product_template pt  
LEFT JOIN ir_translation it  
    ON it.name = 'product.template,field:description'  
    AND it.res_id = pt.id  
    AND it.lang = 'fr_FR'  
WHERE pt.id = 10;
```

Breakdown of the Query:

1. **product_template (pt)** → The main table where the original record exists.
 2. **ir_translation (it)** → Stores translations for specific fields.
 3. **LEFT JOIN with ir_translation:**
 - Filters the translation by **name** (model and field).
 - Matches **res_id** (record ID).
 - Filters by **lang='fr_FR'**.
 4. **COALESCE(it.value, pt.description):**
 - If a translation exists in **ir_translation**, it uses **it.value** (French version).
 - Otherwise, it defaults to the original **description** field.
- While this approach enhances multilingual support, it introduces a **performance cost** due to additional **JOIN** queries.